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COUPLING ASSEMBLY FOR COUPLING A ROD TO A BONE ANCHORING ELEMENT, AND POLYAXIAL BONE ANCHORING DEVICE

Abstract

A coupling assembly for coupling a rod to a bone anchoring element, including a receiving part having first and second ends, a recess at the first end for the rod, and an accommodation space for a head of the bone anchoring element, the accommodation space having an opening at the second end to permit insertion of the head into the receiving part, a retainer element configured to be arranged at least partially in the accommodation space and to hold at least part of the head, and a spring element configured to be arranged at least partially in the accommodation space and to be compressed in an axial direction. The retainer element and the spring element are separate parts. When the retainer element and the spring element are in the accommodation space of the receiving part in a first position, the spring element extends into the recess of the receiving part.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/439,562, filed Feb. 12, 2024, which is a continuation of U.S. patent application Ser. No. 17/376,900, filed Jul. 15, 2021, now U.S. Pat. No. 11,925,391, which is a continuation of U.S. patent application Ser. No. 16/989,366, filed Aug. 10, 2020, now U.S. Pat. No. 11,090,090, which is a continuation of U.S. patent application Ser. No. 15/900,204, filed Feb. 20, 2018, now U.S. Pat. No. 10,779,862, which is a continuation of U.S. patent application Ser. No. 14/596,169, filed Jan. 13, 2015, now U.S. Pat. No. 9,924,971, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/926,683, filed Jan. 13, 2014, the contents of which are hereby incorporated by reference in their entirety, and claims priority from European Patent Application EP 14151009.9, filed Jan. 13, 2014, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

Field

[0002] The invention relates to a coupling assembly for coupling a rod to a bone anchoring element and to a polyaxial bone anchoring device with such a coupling assembly. The coupling assembly includes a receiving part with a channel for receiving the rod, an accommodation space for accommodating a head of a bone anchoring element and a retainer element for retaining the head of the bone anchoring element in the receiving part. Furthermore, a spring element is provided that is biased in such a manner that the retainer element snaps automatically onto the head of the bone anchoring element when the head is being inserted.

Description of Related Art

[0003] From US 2013/0150852 A1, a polyaxial bone anchor including a housing, a bone screw, and a retainer for pivotably coupling the head of the bone screw to the housing is known. The retainer is positioned into the bore of the housing and includes a plurality of alternating tabs and slots circumferentially arranged to define a cavity for receiving the head portion of the bone screw therein. The bone anchor further includes a resilient spring means biasing the retainer towards the lower end of the housing. The head portion of the bone screw may apply a force against the retainer opposing and overcoming the biasing force of the resilient spring means. The resilient spring means may be, for example, a wave washer, a helical spring, an elastomeric member etc. or may be

circumferential or helical slots formed in the retainer.

[0004] U.S. Pat. No. 8,075,603 B2 describes a fastening system consisting of a polyaxial ball and socket joint used in conjunction with a bone screw having threads on one end and a spherical connector on the other end operating as a pivot point about which a connection assembly moves in a polyaxial fashion. A substantially U-shaped connecting assembly has a lower receptacle that operates as a socket for housing an upper retainer ring and a lower split retainer ring. The socket is receptive to the spherical connector which is inserted through the lower split retainer ring causing a momentary displacement thereof which allows for the positioning of the spherical connector between the upper and lower retainer rings. A resilient component, such as two helical springs, positioned between the upper retainer ring and the connecting assembly permits relative predetermined placement and retention of the spherical connector relative to the connector assembly.

SUMMARY

[0005] The above polyaxial bone anchors allow for inserting a spherical head of a bone screw into a receiver by pushing the head against a spring force of a resilient member. However, there is still a need for a coupling assembly and a polyaxial bone anchor with such a coupling assembly that is improved with regard to several aspects, such as the efficiency and safety of the coupling

[0006] It is an object of the invention to provide a coupling assembly for coupling a rod to a bone anchoring element, and a polyaxial bone anchor comprising such a coupling assembly, that provides a safe connection of the bone anchoring element to the coupling assembly with a low insertion force, while also providing a high retention force, and where only a small amount of axial travel or displacement is needed to insert the bone anchoring element into the coupling assembly. Also, the coupling assembly can be easier to manufacture.

[0007] The coupling assembly includes a receiving part with an accommodation space for accommodating the head of the bone anchoring element and a retainer element configured to be positioned at least partially in the accommodation space. Further, the coupling assembly includes a spring element in the form of, for example, a compression spring that is compressible in an axial direction and that fully extends around the central axis in a circumferential direction and that has an axial length so that it can be engaged by an inserted rod. Preferably, the spring element is a wave spring element, which can generate a higher spring force on a given axial length compared to other spring elements. Therefore, the snap-over of the retainer element on the head of the bone anchoring element is facilitated.

[0008] The retainer element further may have at least one horizontal slit at its bottom end that may only need a low insertion force to insert the head into the receiving part. Simultaneously, a retention force that holds the head in the receiving part is high compared to the insertion force. Therefore, the bone anchoring element is effectively prevented from being pulled-out from the lower opening. In addition, because of the small or short insertion path, occurrences of milling under the head or sticking out of the head from the bone can be avoided or reduced.

[0009] The coupling assembly may further include a pressure element for exerting pressure onto the head of the bone anchoring element to lock the bone anchoring element in a specific angular position relative to the receiving part. The retainer element may encompass at least a portion of the pressure element from an outer side thereof, so that increasing a height of the receiving part for accommodating both the retainer element and the pressure element may not be necessary. Hence, a low profile implant can be provided.

[0010] In addition, the receiving part is monolithic and sized such that the retainer element and the spring element, as well as the pressure element, can be mounted from the top opening thereof.

[0011] The pressure element may be held in a position such that the head of the bone anchoring element is held by a frictional force exerted by the pressure element onto the head. The frictional force may be such that the head can still be pivoted by applying a force to overcome the frictional force.

[0012] The coupling assembly can be assembled in situ with a bone anchoring element that has been already inserted into a bone or a vertebra.

[0013] The head of the bone anchoring element may have at its free end an engagement recess for a driver that comprises a groove shaped and arranged in a spiral-like manner. This allows for transmission of high torques onto the bone anchoring element. Furthermore, the free end surface of the anchoring element that has the drive recess may have a spherical shape. In an embodiment where the rod presses directly onto the head without using a pressure element, the spherical shape ensures that a necessary or sufficient contact area between the rod and the head is provided, even at large pivot angles of the bone anchoring element.

[0014] With a coupling assembly according to embodiments of the invention, a modular polyaxial bone anchoring device can be provided that may include several bone anchoring elements that differ with respect to the length of the shank, anchoring features of the shank, such as different thread types and/or thread pitches, different diameters of the shank, and/or with respect to the shank being cannulated or non-cannulated, among other features.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] Further features and advantages of the invention will become more apparent from the description of various embodiments using the accompanying drawings. In the drawings:

[0016] FIG. 1 shows a perspective exploded view of a first embodiment of a bone anchoring device.

[0017] FIG. 2 shows a perspective view of the bone anchoring device of FIG. 1 in an assembled state.

[0018] FIG. 3 shows a perspective view from above of a receiving part according to the first embodiment.

[0019] FIG. 4 shows a perspective view from the bottom of a receiving part shown in FIG. 3.

[0020] FIG. 5 shows a top view of the receiving part shown in FIGS. 3 and 4.

[0021] FIG. 6 shows a cross-sectional view of the receiving part shown in FIGS. 3 to 5, the cross-section being taken along line A-A in FIG. 5.

[0022] FIG. 7 shows a perspective view from above of a retainer element according to the first embodiment.

[0023] FIG. 8 shows a perspective view from a bottom of the retainer element of FIG. 7.

[0024] FIG. 9 shows a top view of the retainer element of FIGS. 7 and 8.

[0025] FIG. 10 shows a cross-sectional view of the retainer element of FIGS. 7 to 9, the cross-section being taken along line B-B in FIG. 9.

[0026] FIG. 11 shows a perspective view from above of a spring element according to the first embodiment.

[0027] FIG. 12 shows a side view of the spring element shown in FIG. 11.

[0028] FIG. 13 shows a top view of the spring element of FIGS. 11 and 12.

[0029] FIGS. 14 and 15 show cross-sectional views illustrating steps of mounting the retainer element to the receiving part according to the first embodiment.

[0030] FIG. 16 shows a cross-sectional view of a fully assembled coupling assembly according to the first embodiment, with the retainer element and the spring element, the cross-section being taken in a direction transverse to an axis of a channel for the rod.

[0031] FIGS. 17 to 18 show cross-sectional views illustrating steps of mounting the coupling assembly of the first embodiment to a bone anchoring element.

[0032] FIG. 19 shows a cross-sectional view of a fully assembled polyaxial bone anchoring device with the coupling assembly according to the first embodiment, with the inserted bone anchoring

element and rod, the cross-section being taken in a plane transverse to an axis of the rod.

[0033] FIG. **20** shows a perspective exploded view of a second embodiment of the polyaxial bone anchoring device, with a second embodiment of the coupling assembly.

[0034] FIG. **21** shows a perspective view from above of a bone anchoring element according to the second embodiment.

[0035] FIG. **22** shows a cross-sectional view of the second embodiment of the polyaxial bone anchoring device, wherein the bone anchoring element is inserted into the coupling assembly, the cross-section being taken in a plane perpendicular to a rod channel axis.

[0036] FIG. **23** shows a cross-sectional view of a fully assembled polyaxial bone anchoring device according to the second embodiment, with an inserted and fixed rod.

[0037] FIG. **24** shows a perspective exploded view of a third embodiment of the polyaxial bone anchoring device, with a third embodiment of the coupling assembly.

[0038] FIG. **25** shows a cross-sectional view of a receiving part according to the third embodiment, the cross-section being taken in a plane perpendicular to a rod channel axis.

[0039] FIG. **26** shows a perspective view from above of a retainer element according to the third embodiment.

[0040] FIG. **27** shows a cross-sectional view of the retainer element shown in FIG. **26**.

[0041] FIG. **28** shows a perspective view from a top of a spring element according to the third embodiment.

[0042] FIG. **29** shows a side view of the spring element of FIG. **28**.

[0043] FIG. **30** shows a perspective view from above of a pressure element according to the third embodiment.

[0044] FIG. **31** shows a perspective view from a bottom of the pressure element shown in FIG. **30**.

[0045] FIG. **32** shows a top view of the pressure element of FIGS. **30** and **31**.

[0046] FIG. **33** shows a cross-sectional view of the pressure element shown in FIGS. **30** to **32**, the cross-section being taken along line D-D in FIG. **32**.

[0047] FIGS. **34** to **35** show cross-sectional views illustrating steps of assembling the coupling assembly according to the third embodiment.

[0048] FIG. **36** shows a cross-sectional view of a step of mounting the coupling assembly of the third embodiment to a bone anchoring element.

[0049] FIG. **37** shows a cross-sectional view of a fully assembled polyaxial bone anchoring device according to the third embodiment.

[0050] FIG. **38** shows a cross-sectional view of the polyaxial bone anchoring device of FIG. **37** with an inserted and fixed rod, wherein the cross-section is taken in a plane transverse to a rod axis.

[0051] FIG. **39** shows a cross-sectional view of the polyaxial bone anchoring device according to the third embodiment, wherein the cross-section is taken in a plane containing the rod axis.

DETAILED DESCRIPTION

[0052] As shown in FIGS. **1** and **2**, a bone anchoring device according to a first embodiment includes a bone anchoring element **1** in the form of a bone screw having a threaded shank **2** and a head **3**. The head **3** has a spherical segment-shaped outer surface portion **3a**, including a greatest outer diameter E of the sphere, and a free end with a recess **3b** for engagement with a screwing-in tool. The bone anchoring device further includes a coupling assembly **4** for coupling a stabilization rod **100** to the bone anchoring element **1**. The coupling assembly **4** includes a receiving part **5**, and a retainer element **6** and a spring element **7** configured to be arranged in the receiving part **5**. A pin **8** may be provided for securing the retainer element **6** against rotation in the receiving part **5**.

[0053] In addition, a locking element **9** in the form of an inner screw is provided for securing the rod **100** in the receiving part **5** and for locking the whole device.

[0054] Referring in particular to FIGS. **3** to **6**, the receiving part **5** is a monolithic part that is substantially cylindrical and has a first end or top end **5a**, a second end or bottom end **5b**, and a central axis of symmetry C passing through the top end **5a** and the bottom end **5b**. A bore **51** is

provided that is coaxial with the central axis C. In a first region adjacent to the top end 5a, the receiving part 5 has a substantially U-shaped recess 52 with a bottom directed towards the bottom end 5b, and two free lateral legs 52a, 52b extending towards the top end 5a. On the legs 52a, 52b, an internal thread 53 is provided that cooperates with the locking element 9. The channel formed by the U-shaped recess 52 is sized so as to receive the rod 100 therein for connecting two or more bone anchoring devices. In a region that extends from a distance above (i.e., closer to the top end 5a than) the bottom of the U-shaped recess 52 to a distance from the bottom end 5b, the bore 51 has a greater diameter compared to other portions of the bore 51 so that an accommodation space 54 is formed to receive the head 3 of the bone anchoring element 1 and to receive the retainer element 6 as well as the spring element 7. The accommodation space 54 has, at its lower end adjacent to the bottom end 5b of the receiving part, a seat portion 54a for the retainer element 6. The seat portion 54a has a smaller diameter than the main portion of the accommodation space 54 and conically tapers toward the bottom end 5b of the receiving part 5. Between the seat portion 54a and the rest of the accommodation space 54, a shoulder 54b is provided that may also function as a stop for the retainer element 6. An upper edge 54c between the accommodation space 54 and the rest of the bore 51 forms a stop for the spring element 7.

[0055] The accommodation space 54 further has an opening 55 at the bottom end 5b, the inner diameter of which is larger than the greatest outer diameter E of the head 3 of the bone anchoring element 1, so that the head 3 can be inserted from the bottom end 5b. The edge or perimeter surrounding the opening 55 has a through-hole 56 that extends into the accommodation space 54 and that serves as a press-fit accommodation of the pin 8. An axis of the through-hole 56 may be parallel to the central axis. The through-hole 56 of the receiving part 5 is positioned in a circumferential direction approximately corresponding to a center of one of the legs 52a, 52b.

[0056] Referring more in detail to FIGS. 7 to 10, the retainer element 6 will be explained. The retainer element 6 resembles a cap-like part, and includes a first end or top end 6a and an opposite second end or bottom end 6b. Adjacent to the bottom end 6b, a spherical segment-shaped recess 61 extends into the retainer element 6 with an inner diameter that matches an outer diameter of the spherical surface portion 3a of the head 3. The extension of the recess 61 in an axial direction is such that when the head 3 is inserted into the recess 61, the bottom end 6b extends below (i.e., closer to the shank 2 than) the region of the head 3 with the greatest diameter E.

[0057] A coaxial bore 62 extends from the top end 6a into the recess 61 to allow access to the head 3 of the anchoring element 1 with a tool. An outer surface of the retainer element 6 has a first portion 63 adjacent to the top end 6a that is substantially conically shaped and that tapers and narrows towards the first end 6a. The first portion 63 may have two opposite flattened sides 63a that facilitate gripping the retainer element 6 with a tool (now shown) for assembling the retainer element 6 and the receiving part 5.

[0058] Approximately in the region with the largest inner diameter of the recess 61, an outwardly protruding annular edge 64 is formed with a maximum outer diameter that is only slightly smaller than an inner diameter of the accommodation section 54 of the receiving part 5, as shown, for example, in FIG. 16. The outwardly protruding annular edge 64 has an upper side 64a that is configured to support the spring element 7. Flattened portions 64b of the annular edge 64 are provided and arranged at substantially 90° with respect to the flattened portions 63a of the first portion 63. On one side at substantially 90° with respect to the flattened portions 64b in a circumferential direction, a substantially U-shaped recess 65 is provided in the outer surface of the annular edge 64 which extends in an axial direction. The recess 65 is configured to receive the pin 8 therein.

[0059] Adjacent to the bottom end 6b, the retainer element 6 has the form of a slit ring 66. The slit ring 66 has a substantially conical outer shape that matches the inner shape of the seat portion 54a of the accommodation space 54. The inner surface of the slit ring 66 forms a portion of the spherical recess 61, so that the inner surface of the slit ring 66 generates a seat for the head 3, to

provide a ball and socket joint between the receiving part 5 and the bone anchoring element 1 when the retainer element 6 is mounted to the receiving part 5. As depicted in FIG. 8, the slit ring 66 is formed by a first vertical slit 66b extending from the bottom end 6b in a substantially vertical direction. From the vertical slit 66b two opposite horizontal slits 66c, 66d extend circumferentially around the central axis C. The horizontal slits 66c, 66d end in widened portions 66e, 66f. Between the end portions 66e, 66f a connecting portion 66g is formed that connects the slit ring 66 to the remainder of the retainer element 6. Hence, the retainer element 6 is a monolithic part including the slit ring 66. The width of the vertical slit 66b and of the horizontal slits 66c, 66d as well as the width of the connecting portion 66g, may be selected such that a desired flexibility of the slit ring 66 is obtained. By means of the slit ring 66, the retainer element 6 is configured to be expanded and compressed in a radial direction. As can be seen in particular in FIGS. 8 and 10, the vertical slit is positioned circumferentially at substantially the same position as the U-shaped recess 65, while the connecting portion 66g is at a circumferentially opposite position.

[0060] As illustrated in FIG. 16, a total axial length of the retainer element 6 when the retainer element 6 is inserted into the receiving part 5 and when the slit ring 66 is seated in the seat portion 54a of the receiving part 5 is such that the top end 6a of the retainer element 6 is approximately at or slightly above the axial height of the bottom of the U-shaped recess 52. It shall be noted that the retainer element 6 acts also as a pressure element on an inserted head 3, as it not only prevents removal of the head 3, but also exerts pressure to the head 3 from the side and from above.

[0061] Referring now to FIGS. 11 to 13, the spring element 7 is formed as a wave spring, and includes substantially circular turns each made out of a flat strip, such as a flat wire. A cross-section of the flat strip may be rectangular. In the embodiment shown, the wave spring has four turns 71, 72, 73, 74. However, in other embodiments, a plurality of turns more or less than four turns may be provided. Each turn has a plurality of wave crests 70a and a plurality of wave troughs 70b that are alternately arranged and that form a ring. The turns 71, 72, 73, 74 that are consecutively stacked in an axial direction around the central axis C are connected to each other, such that adjacent turns offset from one another in a circumferential direction by the length of a crest portion (or a trough portion), as can be seen in particular in FIGS. 11 and 12. This specific shape is known as a crest-to-crest wave spring.

[0062] The turns of the spring element 7 may each be made out of separate flat strips that may be connected to each other, for example, by welding. Alternatively, the spring element can be made out of a single continuous flat strip that is wound around a central axis, so as to form a wave spring shape. Many modifications are possible. The number of turns, i.e. the axial length of the spring element 7, and various other spring properties can be selected such that a desired compression force and return force can be provided by the spring element 7. The axial length of the spring element is selected such that, as can be seen in FIGS. 17 to 19, the spring element 7 can be positioned onto the first portion 63 of the retainer element 6 and fills the accommodation space 54 between the upper surface 64a of the annular edge 64 of the retainer element 6 and the upper edge 54c of the accommodation space 54. An outer diameter of the spring element 7 is slightly smaller than an inner diameter of the accommodation space 54.

[0063] When the spring element 7 is in the accommodation space 54, the spring element 7 may be in a biased condition so that it exerts a pre-load onto the retainer element 6, but can still be compressed further.

[0064] An advantage of a wave spring compared to a helical compression spring is that a wave spring produces the same or similar compression and return force, but can have a shorter axial length than a helical compression spring. As a consequence thereof, the necessary space for the spring element 7 may be smaller or reduced, and/or the travel path for inserting the anchoring element against the spring force can be minimized or reduced.

[0065] The bone anchoring device, as a whole or in part, may be made of one or more bio-compatible materials, such as a bio-compatible metal or a metal alloy, for example titanium,

stainless steel, a nickel-titanium alloy, for example nitinol, or of bio-compatible plastic materials, such as, for example, polyetheretherketon (PEEK), or of a bio-compatible ceramic material. In particular, it may be contemplated that the spring element **7** is made of a superelastic nickel-titanium alloy or of beta titanium.

[0066] The assembly of the coupling assembly will be explained with reference to FIGS. **14** to **16**. First, the pin **8** can be inserted beforehand in the through-hole **56** of the receiving part **5**. Next, the retainer element **6** is inserted into the receiving part **5** from the top end **5a**. The retainer element **6** can be inserted in a 90° tilted manner such that its central axis is oriented at 90° with respect to the central axis C of the receiving part. A less than 90° tilted orientation can also be used for insertion, as illustrated. Then, when a portion of the bottom end **6b** and the annular edge **64** enters the accommodation space **54**, the retainer element **6** can be tilted again, so that its central axis becomes coaxial with the central axis C of the receiving part **5**. The orientation of the retaining element is such that, as depicted in FIG. **15**, the U-shaped axial recess **65** provided at the annular edge **64** engages the pin **8**.

[0067] When the retainer element **6** is seated with the slit ring **66** in the seat portion **54a** of the receiving part **5**, the spring element **7** is inserted into the accommodation space **54** such that the spring element **7** encircles the upper portion **63** of the retainer element **6** and rests on the upper side **64a** of the annular edge **64**. The top end **7a** of the spring element **7** abuts against the stop **54c** provided at the upper side of the accommodating space **54**. In this condition, the spring element **7** is somewhat compressed, so that the spring element **7** exerts a biasing force onto the retainer element **6** to hold the slit ring **66** in the seat **54a**. When the slit ring **66** is placed in to the seat **54a**, the lower edge **6b** of the retainer element **6** can project slightly out of the lower opening **55** of the receiving part **5**. A portion of the spring element **7** extends above the bottom of the channel **52** of the receiving part **5**, so that the spring element **7** can be engaged by the rod **100**.

[0068] Next, as depicted in FIGS. **17** and **18**, the head **3** of the bone anchoring element **1** is inserted through the lower opening **55** into the receiving part **5**. As shown in FIG. **18**, when the head **3** enters the accommodation space **54** through the lower opening **55**, the slit ring **66** is pushed upwards out of the seat **54a**. Simultaneously, the spring element **7** is compressed by the upward movement of the retainer element **6**. In addition, the slit ring **66** is expanded when the head **3** further enters into it. The accommodation space **54** provides space for the expansion after the slit ring **66** is pushed out of the seat **54a**. When the counterforce exerted by the compressed spring element **7** is greater than the necessary force for expanding the slit ring **66** and for sliding the slit ring **66** over the portion of the head **3** having the largest diameter E, the spring force of the compressed spring element **7** causes the slit ring **66** to snap over the head **3**, so that the lower edge **6b** slides over the region of the head **3** with the largest diameter E. The head **3** can be inserted into the retainer element **6** only to such an extent that the head **3** abuts against the upper portion of the spherical recess **61** of the retainer element **6**. The retainer element **6** and the spring element **7** cannot escape through the top end **5a** of the receiving part **5** because of the stop **54c** provided in the accommodation space **54**.

[0069] When the slit ring **66** is positioned below the portion of the head **3** with the greatest diameter E, the head **3** can no longer be pulled-out through the lower opening **55**.

[0070] The slit ring **66** may have a slight undersize with respect to the size of the head **3** when the slit ring **66** is unbiased, such that when the slit ring **66** encompasses the head **3**, a frictional force is exerted onto the head **3**. Hence the head **3** may be held by friction within the receiving part **5**, and the receiving part **5** can be maintained or held temporarily at a specific angular position with respect to the bone anchoring element **1** before locking of the bone anchoring element **1** relative to the receiving part **5**. The spring force exerted by the biased spring element **7** may also contribute to the friction hold of the head **3** in the receiving part **5**.

[0071] As depicted in FIG. **19**, in a next step, the rod **100** is inserted into the U-shaped recess **52** and the locking element **9** is inserted in the receiving part **5**. When the locking element **9** is

tightened, the rod **100** presses onto the top end **6a** of the retaining element **6** and also contacts the spring element **7**. Final tightening of the locking element **9** locks the slit ring **66** of the retaining element **6** and the head **3** in the receiving part **5**. As can be seen in FIG. **19**, in the locked condition, the rod **100** also compresses the spring element **7**.

[0072] In use, the bone anchoring element **1** may be inserted into the bone or into a vertebra prior to mounting the coupling assembly **4**. In an alternative manner of use, the bone anchoring element **1** and the coupling assembly **4** are pre-assembled and thereafter inserted into the bone. A plurality of bone anchoring devices can be connected through a stabilization rod **100**.

[0073] Referring to FIGS. **20** and **21**, a second embodiment of a polyaxial bone anchoring device will be described. The polyaxial bone anchoring device according to the second embodiment differs from the polyaxial bone anchoring device according to the first embodiment in the design of the coupling assembly, and in particular, in the design of the retainer element. All parts that are similar or identical to the previous embodiments are designated with the same reference numerals, and the descriptions thereof will not be repeated.

[0074] The coupling assembly **4'** includes a retainer element **6'**, which lacks the first portion **63** of the retainer element **6** of the first embodiment. The retainer element **6'** includes an annular portion **64'** with a cylindrical inner surface and a slit ring **66** that is identical or similar to the slit ring **66** of the first embodiment. With such a design, the head **3'** of the bone anchoring element **1'** can protrude out of the top end **6a** of the retainer element **6'**. The size of the retainer element **6'** is such that the head **3'** protrudes out of the retainer element **6'** to an extent that allows the rod **100** to press directly onto the head **3'**.

[0075] The bone anchoring element **1'** according to the second embodiment includes a spherical head **3'** that also has a spherical outer surface at a free upper end. In the embodiment, the head as a whole is substantially spherical. The recess **3b'** for the tool may have wings that extend in a spiral-like manner from a center point of the upper free end. In the embodiment shown, the recess **3b'** includes four wings that are formed by a groove **31** that forms the outer contour of a cross with arms each bent in a same direction. By means of this, the engagement surface formed by the recess **3b'** for a tool or driver is enhanced or more robust compared to usual polygon engagements or other recesses or engagements. Therefore, the loads that can be transferred onto the head **3'** are higher. Recesses for drivers and corresponding tools with a similar shape are known, for example, under the trademark Mortorq®. It may be contemplated that similar shapes can be used as drive recesses for the head **3'**.

[0076] As depicted in FIG. **22**, when the head **3'** has entered the receiving part **5** and the retainer element **6'**, the upper portion of the head **3'** including the drive recess **3b'** protrudes out from the top end **6a** of the retainer element **6'**. Also in this embodiment, the head **3** cannot be pulled out through the lower opening **55** once the retainer element **6'** is seated in seat **54a** of the receiving part **5**. The spring element **7** projects into the channel **52** for the rod **100**. Hence, the rod **100** engages the spring element **7** when it is inserted into the recess **52**. When the rod **100** is moved down via the locking element **9**, as depicted in FIG. **23**, the rod **100** compresses the spring element **7**, which in turn urges the retainer element **6'** into the seat **54a** of the receiving part **5**. Finally, the rod **100** presses onto the upper surface of the head **3'**, whereby the head **3'** is pressed into the seat provided by the slit ring **66** of the retainer element **6'**. Because of the spherical shape of the head **3'**, sufficient pressure is exerted onto the head **3'** in both non-pivoted and pivoted states of the bone anchoring element **1'** relative to the receiving part **5**.

[0077] The other steps of mounting the retainer element **6'** to the receiving part **5** are the same as or similar to the first embodiment.

[0078] A third embodiment of a polyaxial bone anchoring device including a coupling assembly will be described with reference to FIGS. **24** to **39**. First, as can be seen in FIG. **24**, the coupling assembly **4''** includes a receiving part **5''**, a retainer element **6''**, a spring element **7''**, and additionally a pressure element **10** for exerting pressure onto the head **3** of the bone anchoring

element **1**. It shall be noted that in this embodiment the bone anchoring element **1** is the same as or similar to the bone anchoring element **1** of the first embodiment. However, embodiments of the invention should not be limited thereto, and another bone anchoring element, for example, that of the second embodiment, can also be used. Parts and portions of the third embodiment that are identical or similar to that of the first or second embodiments are marked with the same reference numerals, and the descriptions thereof will not be repeated.

[0079] The receiving part **5''** of the third embodiment is depicted in a cross-sectional view taken in a place perpendicular to an axis of the rod recess **52** in FIG. **25**. The receiving part **5''** lacks the through-hole **56** for the pin **8** at the bottom end **5b**. For the purpose of holding a pressure element **10** inside the receiving part **5''** and for preventing rotation of the inserted pressure element **10**, the receiving part **5''** includes two transverse pin holes **57a**, **57b** that are positioned in a circumferential direction substantially at centers of the legs **52a**, **52b**. The pin holes **57a**, **57b** are configured to accommodate pins **11a**, **11b** in a press-fit manner. The length of the pins **11a**, **11b** is such that the pins **11a**, **11b** can protrude into the bore **51** of the receiving part **5''** to engage the pressure element **10**. The axial position of the through-holes **57a**, **57b** is such that, as depicted for example in FIG. **36**, the pressure element **10** is held by the pins **11a**, **11b** at its upper end, to be described in greater detail below.

[0080] The retainer element **6''** differs from the retainer elements of the first and the second embodiments in that there is a portion **63''** adjacent to an upper surface **64a** of the annular outwardly protruding edge **64''** that includes a plurality of circumferentially arranged upstanding slightly resilient wall portions **63a** that are separated by axial slots so as to provide elasticity. Each wall portion **63a** has an inwardly protruding edge **63b**. The inwardly protruding edges **63b** are configured to engage a portion of the pressure element **10** so that the pressure element **10** and the retainer element **6''** can be coupled together. An inner and an outer surface of the wall portions **63a** is each substantially conical, tapering and narrowing towards the top end **6a**.

[0081] The annular edge **64''** lacks the recess of the retainer element for receiving the pin in the first embodiment, as there is no pin in the third embodiment for holding the retainer element. The slit ring **66** is the same as or similar to the slit ring **66** from the first and second embodiments.

[0082] As depicted in FIGS. **28** and **29**, the spring element **7''** is also a wave spring, preferably of the type of a crest-to-crest wave spring. The spring element **7''** has three turns in the embodiment shown, but the number of turns is not limited to three, and the spring element in other embodiments can have two or more than three turns, depending on the available space in the accommodation space **54** and on the spring characteristics of the spring element. An inner and an outer diameter of the spring element **7''** is such that the spring element **7''** is configured to be supported on the upper side **64a** of the annular edge **64''** of the retainer element **6''** and to extend around the first portion **63''** having the flexible wall portions **63a**.

[0083] The pressure element **10** has a first end or top end **10a** and a second end or bottom end **10b**. Adjacent to the top end **10a** there is a substantially cylindrical section with a first outer diameter that is only slightly smaller than the inner diameter of the bore **51** of the receiving part **5''** so that the first portion **101** can be slidingly arranged in the bore **51**. Adjacent to the top end **10a** there is also a transverse recess **102** with substantially parallel side walls and a substantially V-shaped bottom **102a** that forms a channel for receiving the rod **100**, where the rod axis is perpendicular to the central axis C. The V-shaped bottom **102a** is configured to support rods of different diameter. By the recess **102**, two upstanding legs **102b**, **102c** are formed that are provided at their top side with smaller U-shaped recesses **103a**, **103b**, respectively, which are open to the top end **10a** and which extend transverse to an axis of the channel. The recesses **103a**, **103b** are configured to receive a portion of the pins **11a**, **11b**, respectively, as can be seen in FIGS. **35** and **36**, for example. Each of the legs **102b**, **102c**, respectively has an outwardly projecting flange **104a**, **104b** that is arranged substantially at the height of the V-shaped bottom **102a** and that extends in a circumferential direction around a portion of each respective leg **102b**, **102c**.

[0084] At either end of the bottom of the channel **102** for the rod **100**, recesses **105a**, **105b** are provided that allow for insertion of the pressure element **10** when the pins **11a**, **11b** are already mounted into the pin holes **57a** and **57b** of the receiving part **5''**.

[0085] The pressure element **10** further includes, adjacent to the bottom end **10b**, a second portion **106** with a smaller diameter compared to the first portion **101**. The second portion **106** is substantially cylindrical and has a spherical segment-shaped recess **107** adjacent to the bottom end **10b**. The spherical recess **107** is configured to exert pressure onto the head **3** of the bone anchoring element **1**. An outwardly protruding edge **108** is provided at the bottom end **10b** for engaging the inwardly protruding edges **63b** of the retainer element **6''** to couple the pressure element **10** to the retainer element **6''**. The edge **108** has a chamfered lower side for facilitating insertion of the second portion **106** into the retainer element **6''**. Furthermore, the pressure element **10** has a coaxial bore **109** for providing access to the head **3** of the bone anchoring element **1** with a tool (not shown).

[0086] The assembly of the coupling assembly **4''** will be explained with reference to FIGS. **34** and **35**. The retainer element **6''** is inserted from the top end **5a** of the receiving part **5''** in a tilted manner. When the retainer element **6''** has reached the accommodation space **54**, the retainer element **6''** is tilted back and placed onto the bottom of the accommodation space **54**, with the slit ring **66** being in the seat **54a**.

[0087] When the retainer element **6''** has been inserted into the receiving part **5''**, the spring element **7''** is also inserted from the top end **5a** of the receiving part **5''** and placed on top of the retainer element **6''** so that the spring element **7''** rests on the annular edge **64''**.

[0088] Next, the pressure element **10** is inserted into the receiving part **5''** from the top end **5a**. As the pins **11a**, **11b** may already be received in the pin holes **57a**, **57b** of the receiving part **5''**, the pressure element **10** can be inserted in a 90° rotated position, so that the recess **102** is oriented at 90° with respect to the U-shaped recess **52** of the receiving part **5''** and the flanges **104a**, **104b** are aligned with the U-shaped recess **52**. In this configuration, the pins **11a**, **11b** can pass through the slits **105a**, **105b**. When the pressure element **10** enters the upper portion **63''** of the retainer element **6''**, the flexible wall portions **63a** are deflected slightly outward to allow the insertion of the bottom end **10b** of the pressure element **10**, until the inwardly protruding edge **63b** of the retainer element **6''** snaps behind or over the outward edge **108** of the lower portion of the pressure element **10**. Once the flanges **104a**, **104b** have entered the accommodation space **54**, the pressure element **10** can be rotated so that the channel **102** of the pressure element **10** and the U-shaped recess **52** of the receiving part **5''** are aligned. Rotation may only be possible when the top end **10a** of the pressure element **10** is below the pins **11a**, **11b**.

[0089] After alignment of the pressure element **10**, the pins **11a**, **11b** are received in the U-shaped recesses **103a**, **103b**, respectively, of the pressure element **10**. The bottom of the U-shaped recesses **103a**, **103b** form an abutment for the pressure element **10** and prevent escaping of the pressure element **10** through the top end **5a** of the receiving part **5''**. In the preassembled state of the coupling assembly **4''**, as depicted in FIG. **35**, the spring element **7''** is slightly pre-compressed.

[0090] The mounting of the coupling assembly **4''** to the bone anchoring element **1** is shown in FIGS. **36** and **37**. As in the first embodiment, the head **3** enters through the lower opening **55** into the retainer element **6''**, and therefore into the accommodation space **54**. The slit ring **66** is pushed out of seat **54a** and the retainer element **6''** is moved upward, whereby the contacting surfaces of the outer rim **108** of the pressure element **10** and of the inwardly protruding edges **63b** of the retainer element **6''** disengage. During upward movement of the retainer element **6''**, the spring element **7''** is compressed, whereby the spring element **7''** abuts against the lower side of the flanges **104a**, **104b** and/or other portions of the lower side of the first portion **101** of the pressure element **10**. The upward movement of the pressure element **10** is limited by the pins **11a**, **11b**.

[0091] During the insertion of the head **3** into the retainer element **6''**, the slit ring **66** expands. As soon as the counterforce exerted by the compressed spring element **7''** is greater than the necessary

force for expanding the slit ring **66** and for sliding the slit ring **66** over the portion of the head **3** with the largest diameter E, the spring force of the compressed spring element **7''** causes the slit ring **66** to snap over the head **3** so that the lower edge **6b** slides over the region with the largest diameter E. The head **3** can be inserted until it abuts against the spherical recess **107** of the pressure element **10**. When the spring force shifts the retainer element **6''** downward, the inwardly protruding edges **63b** and the outward rim **108** engage again.

[0092] The pre-stress exerted by the spring element **7''** and/or a slight undersize of the slit ring **66** compared to the size of the head **3** leads to a frictional hold of the head **3** in the receiving part **5''** before finally locking the head **3**. Also, when the slit ring **66** enters the seat **54a** again, a removal of the head **3** from the receiving part **5''** may no longer be possible.

[0093] Finally, as can be seen in FIGS. **38** and **39**, the rod **100** is inserted and pressed down by tightening the locking element **9** in the receiving part **5''**. The pressure exerted by the pressure element **10** onto the head **3** further presses the slit ring **66** into the seat **54a**, and final tightening locks the head **3** and the retainer element **6''** in the receiving part **5''**.

[0094] Further modifications of the disclosed embodiments may be contemplated. For example, for the bone anchoring element, various different kinds of anchoring elements can be used and combined with the receiving part. The anchoring elements may be, for example, screws with different lengths, screws with different diameters, cannulated screws, screws with different thread forms, nails, hooks, etc. For some anchoring elements, the head and the shank may also be separate parts that can be connected to each other.

[0095] Some possible modifications of the receiving part may include, for example, instead of the U-shaped recess being perpendicular to the central axis, a recess for the rod may be inclined, open to the side, or in the form of a closed channel. Other kinds of locking devices including outer nuts, outer caps, bayonet locking devices, or others can also be utilized. In particular, a two-part locking device that includes a first locking element that exerts pressure via the pressure element onto the head and a second locking element that exerts pressure only onto the rod to lock the head and the rod independently may also be used. In some embodiments, the inner surface portion of the pressure member that contacts the head may not necessarily be spherically-shaped. The inner surface portion may have any other shape that is suitable to exert pressure onto the head. Also, the design of the pressure element can be different and is not limited to the specific design shown in the third embodiment.

[0096] Instead of the pin for retaining the pressure element and for aligning the pressure element with respect to the channel of the receiving part for receiving the rod, other retaining mechanisms can be used.

[0097] In some embodiments, the head of the bone anchoring element is not rotationally symmetric. For example, the head may have two opposite flat surface portions between two spherically-shaped outer surface portions, to achieve pivoting in only one plane.

[0098] Instead of the slit ring, a plurality of vertically extending slits or a combination of substantially vertically and substantially horizontally extending slits may be provided.

[0099] The seat for the slit ring and the outer surface of the slit ring need not be conical. Any shape that provides for safe holding of the slit ring may be contemplated, such as, for example, a spherical shape.

[0100] For the spring element, other spring elements can be used. For example, a helical spring encircling the central axis may be used. In addition, other spring elements like elastomeric cushions may be contemplated.

[0101] It shall be noted that parts of the different embodiments described may also be mixed among each other or exchanged, so that a variety of further embodiments can be generated.

[0102] While the present invention has been described in connection with certain exemplary embodiments, it is to be understood that the invention is not limited to the disclosed embodiments,

but is instead intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims, and equivalents thereof.

Claims

1. A bone anchoring device for coupling a rod to a bone, the bone anchoring device comprising: a bone anchoring element comprising a shank for anchoring to the bone and a head; a receiving part having a first end, a second end below the first end, a central axis extending through the first end and the second end, a recess at the first end for receiving the rod, and an accommodation space at the second end for accommodating the head of the bone anchoring element; a retainer element positionable at least partially in the accommodation space, the retainer element comprising an upwardly facing surface configured to contact and exert an upwardly directed pressure on the head; and a spring separate from the retainer element, the spring comprising a lower surface configured to contact and exert a downwardly directed pressure on the retainer element; wherein when the head, the retainer element, and the spring are held in the receiving part with the head in contact with the upwardly facing surface of the retainer element, the retainer element is configured to remain spaced apart from all portions of the head above a greatest diameter of the head measured in a direction perpendicular to the central axis, and the spring is configured to be blocked from contacting the head by another part of the bone anchoring device.
2. The bone anchoring device of claim 1, wherein the spring is ring-shaped and positionable to extend circumferentially around the central axis in the receiving part.
3. The bone anchoring device of claim 2, wherein the spring is configured to extend uninterruptedly for the entire circumference around the central axis.
4. The bone anchoring device of claim 1, wherein the accommodation space forms an opening at the second end of the receiving part sized to permit insertion of the head of the bone anchoring element therethrough, and the retainer element is radially expandable to facilitate passing of the greatest diameter of the head therethrough.
5. The bone anchoring device of claim 1, wherein the head and/or spring are movable in the receiving part to a position where at least the lower surface of the spring is configured to extend to an axial position lower than the greatest diameter of the head.
6. The bone anchoring device of claim 1, wherein at least part of the spring is movable axially in the receiving part.
7. The bone anchoring device of claim 1, wherein the entire spring is positionable axially higher than the entire retainer element in the receiving part.
8. The bone anchoring device of claim 1, wherein an upper surface of the spring is configured to abut against the pressure element in the receiving part.
9. The bone anchoring device of claim 1, wherein an upper surface of the spring is configured to abut against an inner surface of the receiving part.
10. The bone anchoring device of claim 1, further comprising a pressure element positionable at least partially in the accommodation space and separable from the retainer element and the spring, wherein the pressure element comprises a downwardly facing surface configured to contact and exert a downwardly directed pressure on the head for locking the head in the receiving part.
11. The bone anchoring device of claim 10, wherein the spring is positionable axially between at least part of the retainer element and at least part of the pressure element.
12. The bone anchoring device of claim 10, wherein the spring is configured to be blocked from contacting the head by the pressure element.
13. A bone anchoring device for coupling a rod to a bone, the bone anchoring device comprising: a bone anchoring element comprising a shank for anchoring to the bone and a head; a receiving part having a first end, a second end below the first end, a central axis extending through the first end and the second end, a recess at the first end for receiving the rod, and an accommodation space at

the second end for accommodating the head of the bone anchoring element; a pressure element positionable at least partially in the accommodation space, the pressure element comprising a downwardly facing surface configured to contact and exert a downwardly directed pressure on the head for locking the head in the receiving part; and a spring separate from the pressure element, wherein the spring is configured to extend uninterruptedly around a majority of a circumference around the central axis; wherein when the head, the pressure element, and the spring are held in the receiving part, the spring is configured to be spaced apart from all portions of the head below a greatest diameter of the head measured in a direction perpendicular to the central axis, at least an upper region of the spring is configured to extend to an axial position higher than a bottom of the pressure element, and at least a lower region of the spring is configured to extend to an axial position lower than the bottom of the pressure element.

14. The bone anchoring device of claim 13, wherein the spring is configured to extend uninterruptedly for the entire circumference around the central axis.

15. The bone anchoring device of claim 13, wherein the accommodation space forms an opening at the second end of the receiving part sized to permit insertion of the head of the bone anchoring element therethrough.

16. The bone anchoring device of claim 13, further comprising a retainer element positionable at least partially in the accommodation space, the retainer element comprising an upwardly facing surface configured to contact and exert an upwardly directed pressure on the head.

17. A bone anchoring device for coupling a rod to a bone, the bone anchoring device comprising: a bone anchoring element comprising a shank for anchoring to the bone and a head; a receiving part having a first end, a second end below the first end, a central axis extending through the first end and the second end, a recess at the first end for receiving the rod, and an accommodation space at the second end for accommodating the head of the bone anchoring element; a pressure element positionable at least partially in the accommodation space, the pressure element comprising a downwardly facing surface configured to contact and exert a downwardly directed pressure on the head for locking the head in the receiving part; and a radially expandable element that is separate from the pressure element and that is movable axially in the receiving part; wherein when the head, the pressure element, and the radially expandable element are held in the receiving part, at least part of the radially expandable element is positionable around the pressure element and configured to surround a majority of a circumference around the pressure element in a manner such that the pressure element remains movable axially downwardly relative to both the receiving part and the radially expandable element, while the radially expandable element avoids obstructing the axially downward movement of the pressure element.

18. The bone anchoring device of claim 17, wherein the pressure element is configured to remain spaced apart from all portions of the head below a greatest diameter of the head measured in a direction perpendicular to the central axis.

19. The bone anchoring device of claim 17, wherein the radially expandable element comprises an upwardly facing surface configured to contact and exert an upwardly directed pressure on the head.

20. The bone anchoring device of claim 17, further comprising a spring separate from the pressure element and the radially expandable element, wherein the spring comprises a lower surface configured to contact and exert a downwardly directed pressure on the radially expandable element.
